Software Engineer Assessment – Part 1: Home Task

Nov 10, 2026 · @Caretutors Technologies Ltd

Welcome

Thank you for applying to Caretutors Technologies Ltd. Our assessment has two parts, and this document covers Part 1.

Part	What	Where	Time
Part 1	Build a small web app (this document)	At home	About 2–3 hours, due within 48 hours of receiving this email
Part 2	Practical task and discussion	At our office, in person	50 minutes, date shared after you submit Part 1

If anything is unclear, reply to this email. We will share answers with all candidates so everyone has the same information.

The task: Library Book Manager

Build a small web app where a librarian can manage books and track who has borrowed them, using an AngularJS frontend and a Python Flask backend.

User stories

1. I can see a list of all books and search by title or author.
2. I can filter the list by category (e.g. Fiction, Science, History).
3. I can add a new book and edit an existing one.
4. I can mark a book as borrowed by entering the borrower's name, and mark it returned.
5. I can delete a book.

Business rules

- Title and author are required; published year must be between 1900 and the current year.
- A book that is already borrowed cannot be borrowed again.
- A borrowed book cannot be deleted until it is returned.

Requirements

Backend (Python Flask + SQLite)

Every response, including errors, must be JSON, for example `{"error": "Title is required"}`.

Method	Path	Purpose	Expected responses
GET	/api/books	List books; optional query: search, category	200
POST	/api/books	Add a book	201; 400 on invalid input
GET	/api/books/<id>	One book	200; 404
PUT	/api/books/<id>	Edit a book	200; 400; 404
DELETE	/api/books/<id>	Delete a book	204; 404; 409 if borrowed
POST	/api/books/<id>/borrow	Borrow; body: borrower_name	200; 400; 404; 409 if already borrowed
POST	/api/books/<id>/return	Return	200; 404; 409 if not borrowed

- Book fields: id, title, author, category, published_year, is_borrowed, borrower_name, created_at.
- A seed script that adds 10 sample books.
- Bad input must never cause a 500 error.
- CORS enabled so the frontend can call the API.
- At least 3 pytest tests: add a book, borrow the same book twice (409), invalid input (400).

Frontend (AngularJS 1.8)

Route	Screen	Must show
/books	Book list	Search box, category filter, Borrowed/Available badge, Borrow, Return, Edit and Delete buttons
/books/new and /books/:id/edit	Book form	Title, author, category, year; inline validation; Save and Cancel

- Routing with `ngRoute` (or `ui-router`).
- All HTTP calls in one service (e.g. `BookService`), not inside controllers.
- A loading indicator and a clear error message when an API call fails.
- Ask for the borrower's name before borrowing.
- Any CSS framework (Bootstrap is fine); it should be usable on a phone.

Optional bonus: pagination or sorting, Docker Compose, or one frontend unit test.

How to submit

Reply to this email within 48 hours with a link to your GitHub repository (public, or private with access given to us).

- [ ] Repo with `backend/` and `frontend/` folders
- [ ] README: how to install, seed, run the app and run the tests
- [ ] More than one commit (we look at the history)
- [ ] A short note on what you would improve with more time
- [ ] A note on which parts, if any, you used AI tools for

We care about working features, clean and readable code, sensible API design, validation and clear error messages. AI tools are allowed for Part 1, but you must be able to explain and change every part of your code in person during Part 2.

What to expect in Part 2 (on-site)

After you submit, we will invite you to our office for a 50-minute in-person session that builds on your Part 1 project.

Step	Time	What you will do
1	10 min	Show your Part 1 app and explain how it is built
2	20 min	Add a new feature to your own Part 1 project
3	10 min	Find and fix bugs in a short piece of code
4	10 min	Answer a few AngularJS and Flask questions

- Please bring your own laptop, fully charged, with your charger. It must have Python 3.10+, Node.js, Git, a code editor and a browser installed, and your Part 1 project already set up and running. We will provide Wi-Fi and a power socket.
- Make sure your Part 1 repo can be cloned and run from the README.

- You may use official documentation; AI assistants are not allowed during Part 2.

Good luck, and we look forward to meeting you.